

# Pager Rotations

## CSD380: DevOps

*Module 7: Seeing and Solving Problems + Hypothesis-Driven  
Deployments and Code Reviews*

Assignment 2: Pager Rotation Duties

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# Pager Rotations

- ♦ Why Does On-Call Help?
- ♦ Elements of a Good Rotation
- ♦ What Can Go Wrong?
- ♦ What Should Be Paged?
- ♦ When Pager Load is Too High

# Pager Rotations: Why Does On-Call Help?

- ♦ Having on-call engineers means fast response-times in cases of emergency
- ♦ Handling events promptly increases system reliability and customer trust
- ♦ Can improve regulatory compliance for things like education and healthcare

# Pager Rotations: Elements of a Good Rotation

- ♦ Automation: Software routes issues to the engineer on duty, and provides a single source of truth for who is on-call
- ♦ Ownership: Have a process for issue handoff and escalation so that someone is always responsible for an issue.
- ♦ Documentation: Knowledge bases, runbooks, and issue logs help reduce fatigue and give engineers more time to focus on the more serious problems.
- ♦ Communication: Inevitably, you will wind up solving problems for systems you didn't build. The teams that run those systems are valuable allies in resolving any support issue. Keep the lines of communication open!

# Pager Rotations: What Can Go Wrong?

- ♦ Fatigue: On-call is a vigilant position. Too many low-priority alarms can wear down staff and cause inattention.
- ♦ Unfair Rotations: Either some staff get outsized on-call burdens, or insufficient staff puts everyone on-call for too long. You can't maintain vigilance forever.
- ♦ Unclear Expectations or Insufficient Standards: A high-pressure environment is often not the best time to make important decisions. Not having a playbook of some sort can lead to poor and incomplete fixes.
- ♦ Insufficient Monitoring and Automation: You could have an engineer buckling under call load and not know. You might not even know *who is on duty*.

# Pager Rotations: What Should Be Paged?

Ideally, an on-call engineer should be paged only during an emergency.

If it can be fixed automatically, or if it can wait, don't page.

For example: Automated checks have determined that a customer-facing service is down.

# Pager Rotations: When Pager Load is Too High

Look at your inputs

- Ensure automated alert triggers aren't too aggressive
- Examine rate of bugs entering production
- Was there a change in process? Automation?

# Pager Rotations: When Pager Load is Too High

Look at your process

- ♦ Could you have detected this issue earlier?
- ♦ Was there adequate change control?
- ♦ Can this unique situation inform stress testing in the future?



# Pager Rotations: When Pager Load is Too High

Make sure the remediation is not worse than the problem.

- ♦ Collect a baseline, apply change control, collect metrics on the change
- ♦ Follow up on any process improvements
- ♦ Document, document, document!

# References

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